3 Supervised learning of the truth directions

As mentioned in the introduction, we learn the truth directions from the internal model activations. To clarify precisely how the activations vectors of each model are extracted, we first briefly explain parts of the transformer architecture [Vaswani, 2017, Elhage et al., 2021] underlying LLMs. The input text is first tokenized into a sequence of h tokens, which are then embedded into a high-dimensional space, forming the initial residual stream state $\mathbf{x}_0 \in \mathbb{R}^{h \times d}$, where d is the embedding dimension. This state is updated by L sequential transformer layers, each consisting of a multi-head attention mechanism and a multilayer perceptron. Each transformer layer l takes as input the residual stream activation $\mathbf{x}_{l-1}$ from the previous layer. The output of each transformer layer is added to the residual stream, producing the updated residual stream activation $\mathbf{x}_l$ for the current layer. The activation vector $\mathbf{a}_L \in \mathbb{R}^d$ over the final token of the residual stream state $\mathbf{x}_L \in \mathbb{R}^{h \times d}$ is decoded into the next token distribution.

Following Marks and Tegmark [2023], we feed the LLM one statement at a time and extract the residual stream activation vector $\mathbf{a}_l \in \mathbb{R}^d$ in a fixed layer l over the final token of the input statement. We choose the final token of the input statement because Marks and Tegmark [2023] showed via patching experiments that LLMs encode truth information about the statement above this token. The choice of layer depends on the LLM. For LLaMA3-8B we choose layer 12. This is justified by Figure 2, which shows that true and false statements have the largest separation in this layer, across several datasets.

Following this procedure, we extract an activation vector for each statement s_{ij} in the topic-specific dataset D_i and denote it by $\mathbf{a}_{ij} \in \mathbb{R}^d$, with d being the dimension of the residual stream at layer 12 ($d = 4096$ for LLaMA3-8B). Here, the index i represents a specific dataset, while j denotes an individual statement within each dataset. Computing the LLaMA3-8B activations for all statements (≈ 45000) in all datasets took less than two hours using a single Nvidia Quadro RTX 8000 (48 GB) GPU.

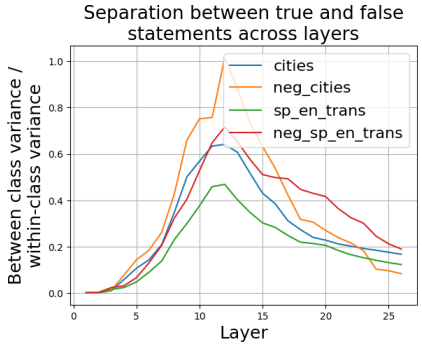

Figure 2: Ratio of the between-class variance and within-class variance of activations corresponding to true and false statements, across residual stream layers, averaged over all dimensions of the respective layer.

As mentioned in the introduction, we demonstrate the existence of *two* truth directions in the activation space: the general truth direction $\mathbf{t}_G$ and the polarity-sensitive truth direction $\mathbf{t}_P$. In Figure 1 we visualise the projections of the activations $\mathbf{a}_{ij}$ onto the 2D subspace spanned by our estimates of the vectors $\mathbf{t}_G$ and $\mathbf{t}_P$. In this visualization of the subspace, we choose the orthonormalized versions of $\mathbf{t}_G$ and $\mathbf{t}_P$ as its basis. We discuss the reasons for this choice of basis for the 2D subspace in Appendix B. The activations correspond to an equal number of affirmative and negated statements from all topic-specific datasets. The top left panel shows both the general truth direction $\mathbf{t}_G$ and the polarity-sensitive truth direction $\mathbf{t}_P$. $\mathbf{t}_G$ consistently points from false to true statements for both affirmative and negated statements and separates them well with an area under the receiver operating characteristic curve (AUROC) of 0.98 (bottom left panel). In contrast, $\mathbf{t}_P$ points from *false to true* for affirmative statements and from *true to false* for negated statements. In the top center panel, we visualise the affirmative truth direction $\mathbf{t}_A$, found by training a linear classifier solely on the activations of affirmative statements. The activations of true and false *affirmative* statements separate along $\mathbf{t}_A$ with a small overlap. However, this direction does not accurately separate true and false *negated* statements (top right panel). $\mathbf{t}_A$ is a linear combination of $\mathbf{t}_G$ and $\mathbf{t}_P$, explaining why it fails to generalize to negated statements.

Now we present a procedure for supervised learning of $\mathbf{t}_G$ and $\mathbf{t}_P$ from the activations of affirmative and negated statements. Each activation vector $\mathbf{a}_{ij}$ is associated with a binary truth label $\tau_{ij} \in \{-1, 1\}$ and a polarity $p_i \in \{-1, 1\}$.

$$\tau_{ij} = \begin{cases} -1 & \text{if the statement } s_{ij} \text{ is } \textit{false} \\ +1 & \text{if the statement } s_{ij} \text{ is } \textit{true} \end{cases} \tag{1}$$

$$p_i = \begin{cases} -1 & \text{if the dataset } D_i \text{ contains } \textit{negated} \text{ statements} \\ +1 & \text{if the dataset } D_i \text{ contains } \textit{affirmative} \text{ statements} \end{cases} \tag{2}$$